

A Guide To SMOKE DETECTORS And FIRE ALARMS



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Residential fires have become a significant problem throughout the country. Fire officials estimate that at least fifty per cent of lives that are lost due to residential fires could be saved by installing early-warning fire-detection devices. The purpose of this guide is to provide information concerning smoke detectors used in conjunction with early-warning smoke- and fire-alarm systems.

Smoke and heat

There are two types of residential fire-detection systems: smoke detectors and heat detectors. The basic residential detection system relies primarily on the use of smoke detectors, which have a sensing chamber, alarm-sounding device and a means of electrical power transmission.

There are two kinds of residential smoke detectors: photoelectric and ionisation. The former are more responsive to fires that begin with a long period of smoldering, whereas the latter are more responsive to flaming fires. Since we cannot predict the type or timing of fire, for maximum protection, it is recommended to use both technologies in homes.

Photoelectric smoke detectors. These aim a light source into a sensing chamber. Smoke produced by a fire affects the intensity of a light beam passing through air. Smoke can block or obscure the beam.

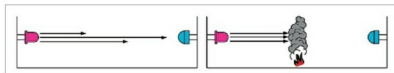


Fig. 1: Light-obscuration smoke detector

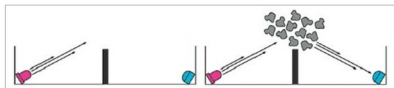


Fig. 2: Light-scattering smoke detector

It can also cause the light to scatter due to reflection off smoke particles. These detectors are designed to sense smoke by utilising these effects of smoke on light.

In a photoelectric light-obscuration smoke detector, when smoke particles partially block the light beam from the inbuilt light source, reduction in light reaching the associated photosensitive device alters its output. Change in this output is sensed by the detector's circuitry, and when the threshold is crossed, an alarm is initiated.

However, in a photoelectric light-scattering smoke detector, a light source is beamed into an area not normally looked out by a photosensitive device. When smoke particles enter the light path, light strikes the particles and is reflected onto the photosensitive device causing the detector to respond.

Photoelectric smoke detectors are better suited to detect slow smoldering fires that are characterised by particulates in 0.4 to 10 micron range.

Ionisation smoke detectors. These use a small radioactive source as a key component for detection of smoke particles. The radionuclide used in these detectors is Americium-241 (Am-241), which is bonded to a metallic foil and sealed within a small chamber. You can find the radioactive button like the one shown in Fig. 4 in most ionisation smoke detectors. A trace of Americium creates charged particles that betray the smoke.

In principle, ionisation smoke detectors have a small amount of radioactive material between two electrically-charged plates, which ionises the air and causes current to flow between the plates. When smoke enters the chamber, it disrupts the flow of ions, thus reducing the flow of current and activating the alarm.

Since reduction in ionised particles results in a decrease in chamber current, that is sensed by the electronic circuitry